Employee Database – SQL Operations

WHERE • ORDER BY • GROUP BY • JOINS • UPDATE • DELETE

Before you begin

Run the creation PDF first. The outputs below assume the original five employees and five salary records are present. Follow the sections in order. UPDATE changes Mamatha's salary and Ravi's city; DELETE removes Bhavani and her salary record.

1. WHERE – filter records

Example A: Employees in department 101 (IT)

```
SELECT * FROM employees WHERE dept_id = 101;
```

emp_id	emp_name	gender	age	dept_id	city
1	Mamatha	Female	21	101	Hyderabad
2	Ravi	Male	22	101	Warangal

Example B: Employees older than 21

```
SELECT * FROM employees WHERE age > 21;
```

emp_id	emp_name	age	city
2	Ravi	22	Warangal
5	Sneha	23	Hyderabad

Example C: Employees from Hyderabad

```
SELECT * FROM employees WHERE city = 'Hyderabad';
```

emp_id	emp_name	city
1	Mamatha	Hyderabad
5	Sneha	Hyderabad

2. ORDER BY – sort records

Example A: Age, ascending (youngest to oldest)

```
SELECT * FROM employees ORDER BY age ASC;
```

emp_name	age
Priya	20
Mamatha	21
Bhavani	21
Ravi	22
Sneha	23

Example B: Age, descending (oldest to youngest)

```
SELECT * FROM employees ORDER BY age DESC;
```

emp_name	age
Sneha	23
Ravi	22
Mamatha	21
Bhavani	21
Priya	20

Example C: Salaries, highest to lowest

```
SELECT * FROM salaries ORDER BY salary DESC;
```

salary_id	emp_id	salary
1	1	50,000
5	5	45,000
3	3	30,000
2	2	25,000
4	4	15,000

3. GROUP BY – group and summarize records

Example A: Count employees in each department

```
SELECT dept_id, COUNT(*) AS total_employees
FROM employees
GROUP BY dept_id;
```

dept_id	total_employees
101	2
102	2
103	1

Example B: Average salary

```
SELECT AVG(salary) AS average_salary FROM salaries;
```

average_salary
33000.00

Example C: Highest salary

```
SELECT MAX(salary) AS highest_salary FROM salaries;
```

highest_salary
50000

4. JOINS – combine related tables

Example A: INNER JOIN employees with departments

```
SELECT employees.emp_name, departments.dept_name
FROM employees
INNER JOIN departments
ON employees.dept_id = departments.dept_id;
```

emp_name	dept_name
Mamatha	IT
Ravi	IT
Priya	HR
Bhavani	Non-IT
Sneha	HR

Example B: LEFT JOIN employees with departments

```
SELECT employees.emp_name, departments.dept_name
FROM employees
LEFT JOIN departments
ON employees.dept_id = departments.dept_id;
```

emp_name	dept_name
Mamatha	IT
Ravi	IT

emp_name	dept_name
Priya	HR
Bhavani	Non-IT
Sneha	HR

Note: This sample has a matching department for every employee, so INNER JOIN and LEFT JOIN show the same five rows. LEFT JOIN would also keep an employee if their department had no match.

Example C: Join all three tables

```
SELECT employees.emp_name, departments.dept_name, salaries.salary
FROM employees
INNER JOIN departments
ON employees.dept_id = departments.dept_id
INNER JOIN salaries
ON employees.emp_id = salaries.emp_id;
```

emp_name	dept_name	salary
Mamatha	IT	50,000
Ravi	IT	25,000
Priya	HR	30,000
Bhavani	Non-IT	15,000
Sneha	HR	45,000

5. UPDATE – modify existing records

Example A: Change Mamatha's salary from 50,000 to 55,000

```
UPDATE salaries
SET salary = 55000
WHERE emp_id = 1;

SELECT * FROM salaries WHERE emp_id = 1;
```

salary_id	emp_id	salary
1	1	55,000

Example B: Change Ravi's city to Hyderabad

```
UPDATE employees
SET city = 'Hyderabad'
WHERE emp_id = 2;

SELECT * FROM employees WHERE emp_id = 2;
```

emp_id	emp_name	gender	age	dept_id	city
2	Ravi	Male	22	101	Hyderabad

6. DELETE – remove records

Because salaries.emp_id references employees.emp_id, delete the salary row first, then delete the employee row.

```
-- Step 1: Delete Bhavani's salary record
DELETE FROM salaries
WHERE emp_id = 4;

-- Step 2: Delete Bhavani's employee record
DELETE FROM employees
WHERE emp_id = 4;

-- Step 3: Display remaining employees
SELECT * FROM employees;
```

emp_id	emp_name	gender	age	dept_id	city
1	Mamatha	Female	21	101	Hyderabad
2	Ravi	Male	22	101	Hyderabad
3	Priya	Female	20	102	Karimnagar
5	Sneha	Female	23	102	Hyderabad

Quick revision

Command	Purpose
WHERE	Filter records using a condition
ORDER BY	Sort records in ascending or descending order
GROUP BY	Group records, often with COUNT, SUM or AVG
JOIN	Combine related data from tables
UPDATE	Change existing values

Command	Purpose
DELETE	Remove selected rows

Safety reminder: UPDATE or DELETE without a WHERE clause can affect every row in the table. Check the condition carefully before running these commands.